

Marthan Lanuzga

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EDUCATION

University of the Philippines Diliman

Master of Engineering in Artificial Intelligence

Quezon City, PH

2026 – 2028 (Expected)

- **Relevant Coursework:** Artificial Intelligence, Classical Machine Learning, Advanced Machine Learning, Computational Linear Algebra for AI, and Computer Vision.
- Planned focus areas include Computer Vision, Autonomous Systems, and Machine Learning Operations (MLOps).

Catanduanes State University

BS Computer Engineering - GWA: 1.78

Catanduanes, PH

2022 – 2026

- **Relevant Coursework:** Computer Networks & TCP/IP, Data Structures & Algorithms, OOP, Machine Learning, Edge AI, Embedded Systems, Digital Signal Processing.

EXPERIENCE

DOST – Advanced Science and Technology Institute

Software Engineering Intern

Quezon City, PH

May 2025 – August 2025

- **Optimized ML pipelines** for large-scale computer vision projects, reducing manual feature extraction by 60% while ensuring 100% compliance with ISO-compliant government thresholds.
- Developed React front-end interfaces with underlying machine learning models alongside a **9-member Agile team**, enabling real-time data visualization.
- Refactored machine learning pipeline configurations to replace hardcoded parameter sets with dynamic, validated, and standard operational thresholds to guarantee strict data integrity.

Sasquatch

Lead Software Developer

Caloocan City, PH

September 2025 – Present

- Architected a behavior-aware AI assistant using deterministic static analysis and a tiered LLM escalation layer that **eliminated 95%** of common syntax and type errors during the development phase via local deterministic analysis, maintaining a **sub50ms execution overhead**.
- Designed a **deterministic-first architecture** with clear separation between local analysis and a future **AI escalation layer**.
- Built a scalable memory and autonomy framework using **SQLite**, enabling long-term context retention, conversational continuity, and adaptive agent behavior across sessions.

PROJECTS

AKMA | Autonomous Knowledge Architecture

May 2026 – June 2026

- Engineered adversarial **multi-stage validation logic**, including a Lazy Delta layer, achieving a **100% rejection rate** against poisoned or unverified web claims.
- Designed a multi-agent Retrieval-Augmented Generation (**RAG**) pipeline using Python and LLM APIs to **autonomously expand knowledge bases** without manual curation.
- Authored the peer-reviewed-ready technical manuscript, "AKMA: Autonomous Knowledge Mutation for Persistent RAG Systems," publicly archived and persistently indexed under **DOI: 10.5281/zenodo.20668815**.

Pathfinder | Autonomous Tourism Agent

Full-Stack AI Developer

September 2025 – October 2025

- Created a **greedy heuristic algorithm** to approximate the **Traveling Salesperson Problem (TSP)** on edge devices, clustering destinations geographically while adhering to **O(n) complexity constraints**.
- Designed an offline distributed architecture deployed on Raspberry Pi hardware, achieving **90% system availability** and enabling query processing in **zero-connectivity environments**.
- Reduced RAG pipeline latency by **90% (~300ms)** by architecting a semantic caching layer on ChromaDB, eliminating redundant vector computations and cutting LLM operational costs by **~80%**.

- **Developed a CNN in TensorFlow/Keras** to classify drug names and dosages from raw pill bottle images, serving as the core classification layer for downstream OCR and NER processing.
- Trained a custom **NER model** using **spaCy** to extract pharmacological entities, cross-referencing outputs against the **OpenFDA API** to ensure **100% data compliance**.
- Architected a **hybrid computer vision pipeline** using **OpenCV** for adaptive image thresholding and **PaddleOCR** for text extraction on high-noise medical imagery.

PokeBind | E-Commerce Platform

March 2026.

- Built REST API routes using Next.js API routes / Node.js-style backend logic for order validation, payment handling, and inventory updates.
- Integrated the **PayMongo API** to handle **secure local payment processing**, including **GCash and Maya** support via specialized webhook listeners.
- Developed responsive UI using **Tailwind CSS**, implementing server-side logic for order validation and **automated stock restocking** via Next.js API routes.

HONORS AND AWARDS

1st Place, Team Leader — 2026 Regional ICpEP Robothon Challenge

- Led an SDG-themed robotics project, overseeing design, programming, testing, and team coordination.

Regional Representative — 2023 Regional ICpEP Robothon Challenge

- Represented the institution in an SDG-themed robotics competition, contributing to system development, programming, and hardware integration.

SKILLS

Languages: Python, TypeScript, C++, JavaScript, SQL, HTML/CSS.

Web Development: Next.js, React, Vue, FastAPI, Supabase, Tailwind CSS.

Backend: Node.js, Express.js, FastAPI, REST APIs.

AI & Data Science: PyTorch, TensorFlow, Keras, RAG, OpenCV, Pandas, NumPy, Scikit-Learn.

Databases: PostgreSQL, Supabase, MongoDB, MySQL, SQLite, ChromaDB, FAISS. **Tools:** Git, GitHub, GitHub Actions, Docker, Jest, API debugging, DBeaver.